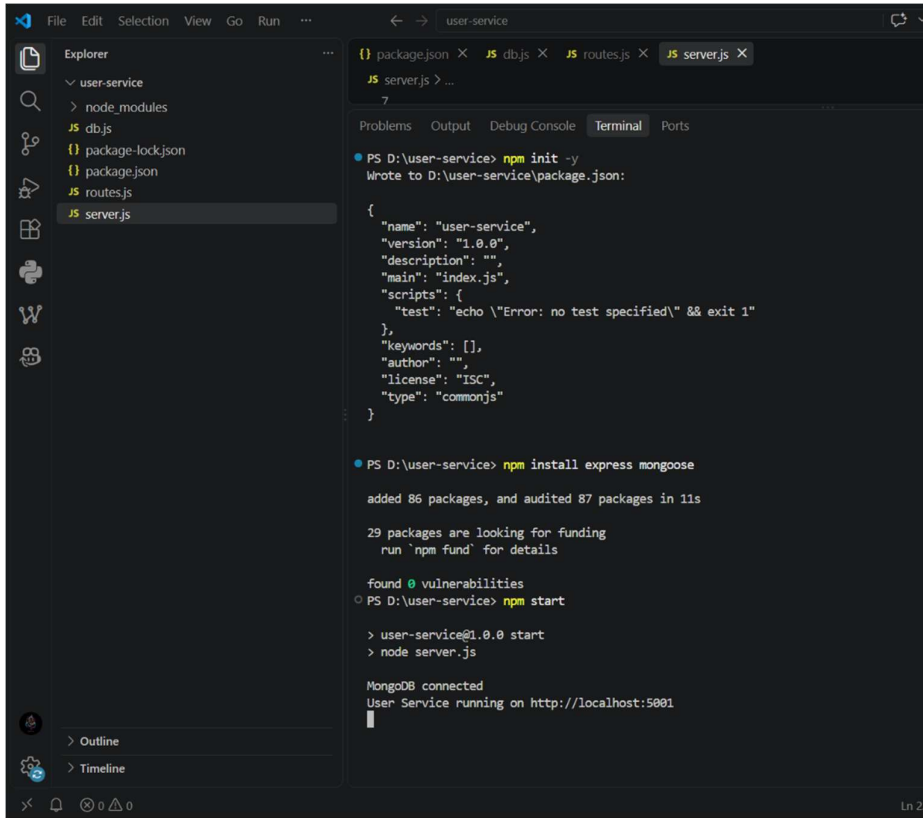




```
File Edit Selection View Go Run ... user-service
{} package.json X JS db.js X JS routes.js X JS server.js X
JS server.js > ...
7
Problems Output Debug Console Terminal Ports
• PS D:\user-service> npm init -y
Wrote to D:\user-service\package.json:

{
  "name": "user-service",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

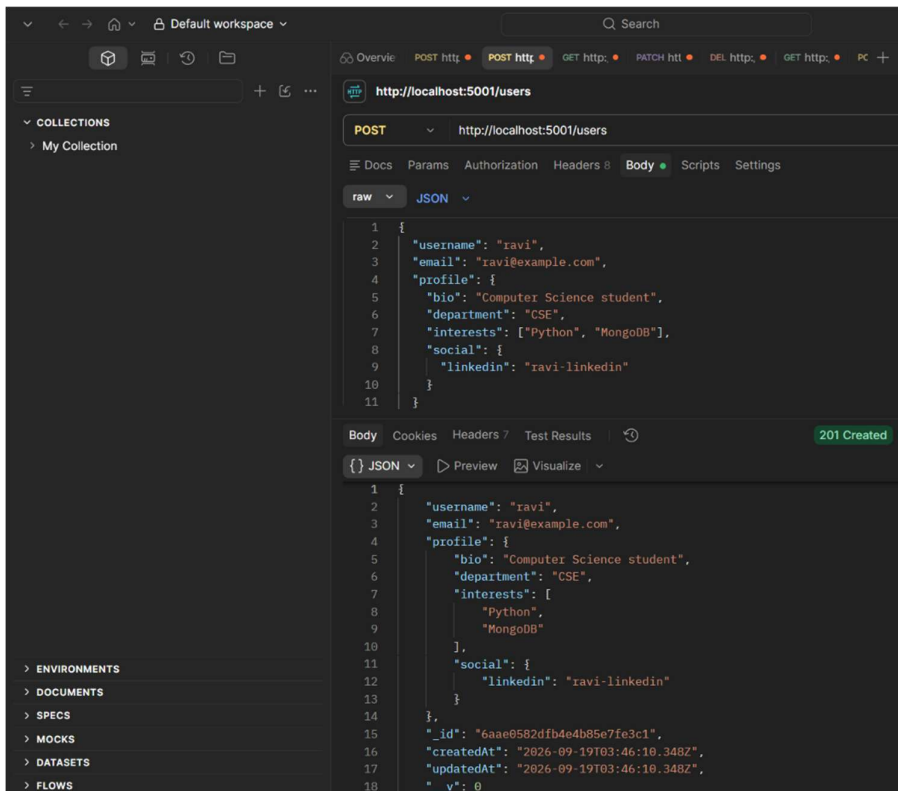
• PS D:\user-service> npm install express mongoose
added 86 packages, and audited 87 packages in 11s

29 packages are looking for funding
run 'npm fund' for details

found 0 vulnerabilities
○ PS D:\user-service> npm start

> user-service@1.0.0 start
> node server.js

MongoDB connected
User Service running on http://localhost:5001
Ln 22
```



```
Default workspace
Search
Overview POST http GET http PATCH http DEL http GET http PC +
http://localhost:5001/users
POST http://localhost:5001/users
Docs Params Authorization Headers 8 Body Scripts Settings
raw JSON
1 {
2   "username": "ravi",
3   "email": "ravi@example.com",
4   "profile": {
5     "bio": "Computer Science student",
6     "department": "CSE",
7     "interests": ["Python", "MongoDB"],
8     "social": {
9       "linkedin": "ravi-linkedin"
10    }
11  }
12 }
Body Cookies Headers 7 Test Results 201 Created
{} JSON Preview Visualize
1 {
2   "username": "ravi",
3   "email": "ravi@example.com",
4   "profile": {
5     "bio": "Computer Science student",
6     "department": "CSE",
7     "interests": [
8     "Python",
9     "MongoDB"
10    ],
11   "social": {
12     "linkedin": "ravi-linkedin"
13   }
14 },
15   "_id": "6aae0582dfb4e4b85e7fe3c1",
16   "createdAt": "2026-09-19T03:46:10.348Z",
17   "updatedAt": "2026-09-19T03:46:10.348Z",
18   "__v": 0
19 }
```

Search

Overview POST http: POST http: GET http: PATCH htt DEL http: GET http: POS

... HTTP http://localhost:5001/users

POST http://localhost:5001/users

Docs Params Authorization Headers 8 Body Scripts Settings

raw JSON

```
1 {
2   "username": "anita",
3   "email": "anita@example.com",
4   "profile": {
5     "bio": "Backend learner",
6     "phone": "9876543210"
7   }
8 }
```

Body Cookies Headers 7 Test Results 201 Crea

{ } JSON Preview Visualize

```
1 {
2   "username": "anita",
3   "email": "anita@example.com",
4   "profile": {
5     "bio": "Backend learner",
6     "phone": "9876543210"
7   },
8   "_id": "6aae30d9022f09d60d69bea7",
9   "createdAt": "2026-09-19T06:51:05.952Z",
10  "updatedAt": "2026-09-19T06:51:05.952Z",
11  "__v": 0
12 }
```

Q Search

Overview POST http: GET http: GET http: PATCH htt DEL http: GET http: POS

HTTP http://localhost:5001/users/6aae0582dfb4e4b85e7fe3c1

GET http://localhost:5001/users/6aae0582dfb4e4b85e7fe3c1

Docs Params Authorization Headers 8 Body Scripts Settings

Query Params

Key	Value	Description
Key	Value	Description

Body Cookies Headers 7 Test Results 200 OK

{ } JSON Preview Visualize

```
1 {
2   "_id": "6aae0582dfb4e4b85e7fe3c1",
3   "username": "ravi",
4   "email": "ravi@example.com",
5   "profile": {
6     "bio": "Computer Science student",
7     "department": "CSE",
8     "interests": [
9       "Python",
10      "MongoDB"
11    ],
12    "social": {
13      "linkedin": "ravi-linkedin"
14    }
15  },
16  "createdAt": "2026-09-19T03:46:10.348Z",
17  "updatedAt": "2026-09-19T03:46:10.348Z",
18  "__v": 0
19 }
```

Standing by for launch

space ▾

Search

Overview POST http: POST http: GET http: PATCH htt: DEL http: GET http: POS + ▾

HTTP http://localhost:5001/activities

POST ▾ http://localhost:5001/activities

Docs Params Authorization Headers 8 Body ● Scripts Settings

raw ▾ JSON ▾

```
1 {
2   "user_id": "6aae0582dfb4e4b85e7fe3c1",
3   "action": "VIEW_BOOK"
4 }
```

Body Cookies Headers 7 Test Results | ↺

201 Created

{ } JSON ▾ ▸ Preview 🖼 Visualize ▾

```
1 {
2   "user_id": "6aae0582dfb4e4b85e7fe3c1",
3   "action": "VIEW_BOOK",
4   "_id": "6aae32ae022f09d60d69beac",
5   "timestamp": "2026-09-19T06:58:54.649Z",
6   "__v": 0
7 }
```

HTTP http://localhost:5001/activities

POST ▾ http://localhost:5001/activities

Docs Params Authorization Headers 8 Body ● Scripts Settings

raw ▾ JSON ▾

```
1 {
2   "user_id": "6aae0582dfb4e4b85e7fe3c1",
3   "action": "SEARCH_BOOK"
4 }
```

Body Cookies Headers 7 Test Results | ↺

{ } JSON ▾ ▸ Preview 🖼 Visualize ▾

```
1 {
2   "user_id": "6aae0582dfb4e4b85e7fe3c1",
3   "action": "SEARCH_BOOK",
4   "_id": "6aae32ed022f09d60d69bead",
5   "timestamp": "2026-09-19T06:59:57.565Z",
6   "__v": 0
7 }
```

Search

Overview POST http: POST http: GET http: PATCH htt: DEL http: GET http: POS

http://localhost:5001/activities

POST http://localhost:5001/activities

Docs Params Authorization Headers 8 Body Scripts Settings

raw JSON

```
1 {
2   "user_id": "6aae0582dfb4e4b85e7fe3c1",
3   "action": "LOGIN"
4 }
```

Body Cookies Headers 7 Test Results 201 Cre

{ } JSON Preview Visualize

```
1 {
2   "user_id": "6aae0582dfb4e4b85e7fe3c1",
3   "action": "LOGIN",
4   "_id": "6aae32fb022f09d60d69beae",
5   "timestamp": "2026-09-19T07:00:11.392Z",
6   "__v": 0
7 }
```

Overview POST http: POST http: GET http: PATCH htt: DEL http: GET htt

http://localhost:5001/activities

POST http://localhost:5001/activities

Docs Params Authorization Headers 8 Body Scripts Settings

raw JSON

```
1 {
2   "user_id": "6aae0582dfb4e4b85e7fe3c1",
3   "action": "BORROW_BOOK"
4 }
```

Body Cookies Headers 7 Test Results

{ } JSON Preview Visualize

```
1 {
2   "user_id": "6aae0582dfb4e4b85e7fe3c1",
3   "action": "BORROW_BOOK",
4   "_id": "6aae3309022f09d60d69beaf",
5   "timestamp": "2026-09-19T07:00:25.262Z",
6   "__v": 0
7 }
```

Q Search

Overview POST http: GET http: GET http: PATCH htt: DEL http: GET http: POS +

http://localhost:5001/activities/6aae0582dfb4e4b85e7fe3c1

GET http://localhost:5001/activities/6aae0582dfb4e4b85e7fe3c1

Docs Params Authorization Headers 6 Body Scripts Settings

Body Cookies Headers 7 Test Results 200 OK

{ } JSON Preview Visualize

```
1  [
2      {
3          "_id": "6aae3309022f09d60d69beaf",
4          "user_id": "6aae0582dfb4e4b85e7fe3c1",
5          "action": "BORROW_BOOK",
6          "timestamp": "2026-09-19T07:00:25.262Z",
7          "__v": 0
8      },
9      {
10         "_id": "6aae32fb022f09d60d69beae",
11         "user_id": "6aae0582dfb4e4b85e7fe3c1",
12         "action": "LOGIN",
13         "timestamp": "2026-09-19T07:00:11.392Z",
14         "__v": 0
15     },
16     {
17         "_id": "6aae32ed022f09d60d69bead",
18         "user_id": "6aae0582dfb4e4b85e7fe3c1",
19         "action": "SEARCH_BOOK",
20         "timestamp": "2026-09-19T06:59:57.565Z",
21         "__v": 0
22     },
23     {
24         "_id": "6aae32ae022f09d60d69beac",
25         "user_id": "6aae0582dfb4e4b85e7fe3c1",
26         "action": "VIEW_BOOK",
27         "timestamp": "2026-09-19T06:58:54.649Z",
28         "__v": 0
29     }
30 ]
```

✦ Standing by for launch

MongoDB Compass - localhost:27017/user_service_db.users

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

- localhost:27017
 - BookFlow
 - addressesDB
 - admin
 - boxOffice
 - config
 - contactData
 - empinfo
 - financialData
 - local
 - movie
 - people
 - student
 - student_records
 - tvShows
 - user
 - user_service_db
 - activities

localhost:27017 > user_service_db > users

Documents 2 Aggregations Schema Indexes 2 Validation

Type a query: { field: 'value' } or [Generate query](#)

EXPLAIN RESET FIND

ADD DATA BULK EXPORT DATA EXPORT CODE

25 1 - 2 of 2

```
{
  "_id": ObjectId('6aae9582dfb4e4b85e7fe3c1'),
  "username": "ravi",
  "email": "ravi@example.com",
  "profile": Object (4)
  createdAt: ISODate('2026-09-19T03:46:10.348+00:00') September 19, 2026 at 3:46:10 AM UTC
  updatedAt: ISODate('2026-09-19T03:46:10.348+00:00') September 19, 2026 at 3:46:10 AM UTC
  __v: 0
}
```

```
{
  "_id": ObjectId('6aae30d9022f09d60d69bea7'),
  "username": "anita",
  "email": "anita@example.com",
  "profile": Object (2)
  createdAt: ISODate('2026-09-19T06:51:05.952+00:00') September 19, 2026 at 6:51:05 AM UTC
  updatedAt: ISODate('2026-09-19T06:51:05.952+00:00') September 19, 2026 at 6:51:05 AM UTC
  __v: 0
}
```

Compass 1.50.0 is downloading

MongoDB Compass - localhost:27017/user_service_db.activities

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

- localhost:27017
 - BookFlow
 - addressesDB
 - admin
 - boxOffice
 - config
 - contactData
 - empinfo
 - financialData
 - local
 - movie
 - people
 - student
 - student_records
 - tvShows
 - user
 - user_service_db
 - activities

localhost:27017 > user_service_db > activities

Documents 4 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

EXPLAIN RESET FIND

ADD DATA BULK EXPORT DATA EXPORT CODE

25 1 - 4 of 4

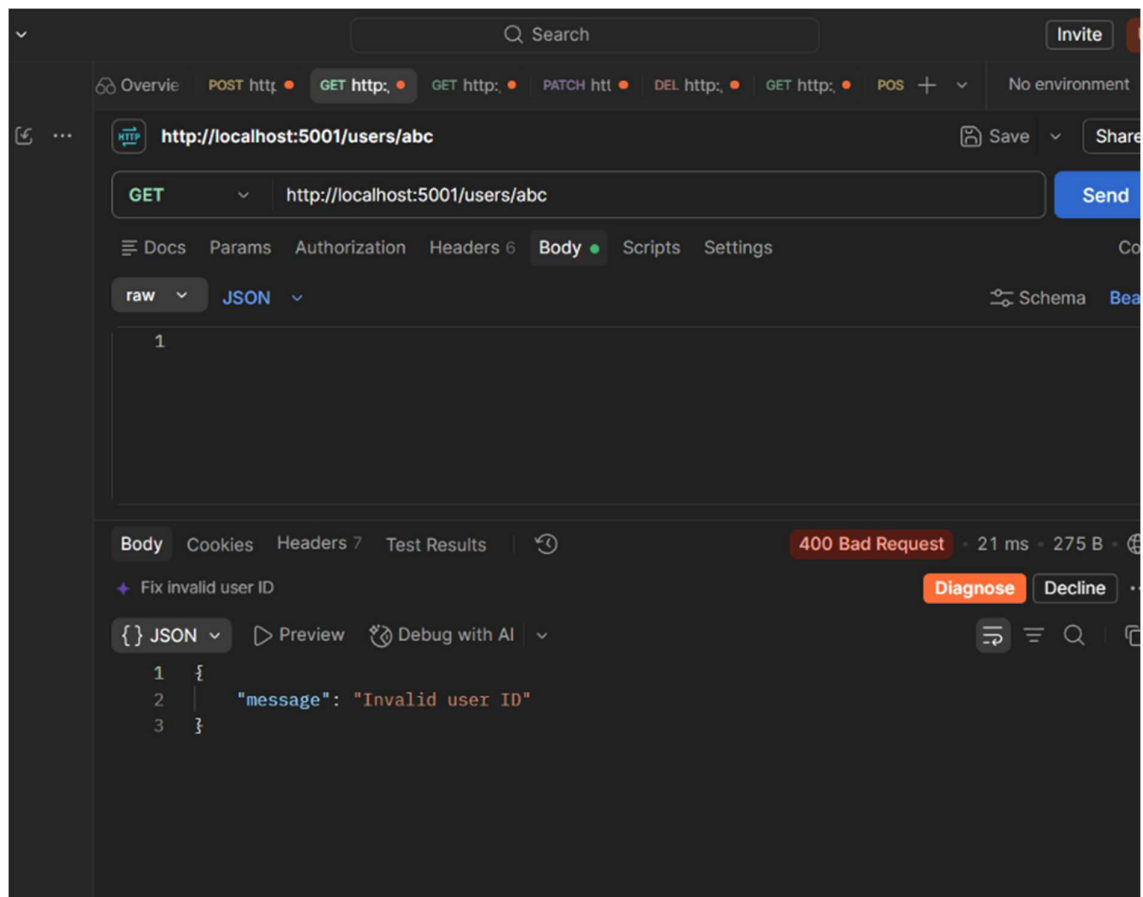
```
{
  "_id": ObjectId('6aae32ae022f09d60d69beac'),
  "user_id": ObjectId('6aae9582dfb4e4b85e7fe3c1'),
  "action": "VIEW_BOOK",
  "timestamp": ISODate('2026-09-19T06:58:54.649+00:00') September 19, 2026 at 6:58:54 AM UTC
  __v: 0
}
```

```
{
  "_id": ObjectId('6aae32ed022f09d60d69bead'),
  "user_id": ObjectId('6aae9582dfb4e4b85e7fe3c1'),
  "action": "SEARCH_BOOK",
  "timestamp": ISODate('2026-09-19T06:59:57.565+00:00') September 19, 2026 at 6:59:57 AM UTC
  __v: 0
}
```

```
{
  "_id": ObjectId('6aae32fb022f09d60d69beae'),
  "user_id": ObjectId('6aae9582dfb4e4b85e7fe3c1'),
  "action": "LOGIN",
  "timestamp": ISODate('2026-09-19T07:00:11.392+00:00') September 19, 2026 at 7:00:11 AM UTC
  __v: 0
}
```

```
{
  "_id": ObjectId('6aae3309022f09d60d69beaf'),
  "user_id": ObjectId('6aae9582dfb4e4b85e7fe3c1'),
  "action": "BORROW_BOOK",
  "timestamp": ISODate('2026-09-19T07:00:25.262+00:00') September 19, 2026 at 7:00:25 AM UTC
  __v: 0
}
```

Compass 1.50.0 is downloading



Conclusion:

The User Profiles and Activity Logging Service was successfully implemented using Node.js, Express.js, MongoDB, and Mongoose. The service provides REST API endpoints for creating and retrieving user profiles and activity logs. MongoDB stores the users and activities in separate collections, while Mongoose provides schema and model management. The profile field uses a flexible structure, allowing different users to have different profile fields. This service runs independently on port 5001 and is separate from the FastAPI BookFlow service used in Experiment 2.